

AIR UNIVERSITY
Department of Electrical
College of Engineering
ARTIFICIAL INTELLIGENCE

LAB01

Lab Objective: Introduction to Python and Cloud Computing

Class ID: _____

LabDate: _____

Student's ID: _____

Lab 01	Introduction to Python and Cloud Computing				
PLO's	PLO5 – Modern Tool Usage	BLOOM'S TAXONOMY		P2 – Set	
LAB TASK PERFORMANCE					
CLO's	Aspects of Assessments	Excellent (75-100%)	Average (50-75%)	Poor ($\leq 50\%$)	M
CLO1 100%	<u>Tools Utilization</u> Apply and discover different basic level functions of python.	Properly and accurately implement the functions and modules of python as per given task.	Implement the functions and modules of python with minor errors that give output with slightly differ from the given task.	Does not know how to implement the functions and modules and achieved the required output as describe in a task.	
Total Marks: 10					

Graded by Engineer: _____

Date: _____

Remarks: _____

Objective:

To make students familiar with the concept of cloud computing. Hands-on activity on Python in the Google Colab/ Spyder IDE environment.

Google Colab:

Google Colab is a free cloud service and now it supports free GPU! You can; improve your Python programming language coding skills. develop deep learning applications using popular libraries such as Keras, TensorFlow, PyTorch, and OpenCV.

Spyder IDE:

Spyder is a IDE for python programming, where it offers help to programmers to identify list of functions available under an object (python is an object oriented language) or number and data type of parameters accepted by a function. Like Google Colab you can make use of popular libraries such as Keras, TensorFlow, PyTorch, and OpenCV.

Python

Python is a general-purpose and open source coding **language**—which means that, unlike HTML, CSS, and JavaScript, it can be **used** for other types of **programming** and software development besides web development. That includes back end development, software development, data science and writing system scripts among other things. It is one of the most popular language in AI programming and model development.

A number of packages are available in Python for efficient implementation of AI algorithms. One of the most extensively used package is called as “numpy”. Numpy is primarily used for mathematical operations.

Numpy Primer:

Let's do some hands-on experiments:

```
>>import numpy as np
>>A=np.array([[2 ,3],[ 2,2]]) #define a matrix#
>>AI=np.linalg.inv(A) #calculate inverse
```

To check your variables: To see the value of a particular variable, move your mouse to the variable, a pull down menu will appear you can click ‘view’ to see the value of that variable.

To remove any variable say A in python console use command

```
>>del A
```

for multiple variables use >>del A, AI

To access elements of an array we use square brackets >> A[0,0]

Indexing starts from 0 in python

To know dimensions of a variable `b` `>>> np.shape(b)` # will return rows and cols

To know type of variable `b` `>>> type(b)`

In above example variable `b` is a tuple. Tuple is a data type in python which means a sequence i.e a collection of numbers, names etc separated by a comma. Note that a tuple is not an array

```
b=np.ones((2,3))  
  
bshape=np.shape(b)  
  
type(bshape)
```

Now the variable 'bshape' is a tuple, length of bshape can be found using `>>> len(bshape)`, which will be '2' as there are total 2 elements in the sequence . consequently the rows and columns returned by `np.shape` commands are not integers and therefore cannot be used as numbers. To find the dimensions in integer form:

`>>>len(b)` ## will give number of rows

`>>>len(b[0])` ## will give number of columns

Note that lists and tuples both are sequence of items. However lists are mutable i.e. there particular item can be changed whereas tuples are immutable.

Lists can be formed by using square brackets `>>> c=['apples','oranges','berries']`

To change a particular item `>>>c[0]='mangoes'`

Tuples can be formed using paranthesis `>>> d=('apples','oranges','berries')`

For accessing all rows of a particular column (or vise versa) of an array:

```
A[:,1]] # will return all rows of column 1  
A[:,1,2]] # will return all entries of column 1 and 2
```

Creating all zeros and all ones

`>>>np.zeros((2,3))` # will create a 2x3 array of all zeros

`>>>np.ones((2,3))` # will create a 2x3 array of all ones

We can fix the data type as well

`>>>np.ones((2,3),dtype=np.int16)` # will create a 2x3 array of all ones

Multiplication of arrays:

`>>>C=np.ones((3,3))`

`>>>C*C` ## this will not be matrix multiplication, rather an element-wise multiplication. For matrix multiplication you may use `>>> np.matmul()`

For transposition

>>a.T

Lab Task:

(CLO1, Marks 4) Complete hands-on activities as prescribed above.

(CLO 6, Marks 1) Perform properly handling of lab.

Home Task:

For all tasks given below, submit the relevant commands and the corresponding output values in a word file/hardcopy with your name and ID.

1. (CLO1, Marks 2) Generate a 2x2 numpy array `[[2,3],[2,2]]` and call it A. Now using the concept of broadcasting add 2 to each entry of A. Call the new matrix B. (Hint: Search 'Broadcasting in Python')
2. (CLO1, Marks 1) Perform matrix multiplication of A and B, call the output E.
3. (CLO1, Marks 1) Add the rows of E so that the output is a 1x2 vector 'F'.
4. (CLO1, Marks 1) Add the columns of E so that the output is a 2x1 vector 'G'